
Pregunta

a) (1,25 punts) Calcula les integrals indefinides $\int \frac{3}{x-1} dx$ i $\int \frac{2}{(x+1)^2} dx$.

Solució. $\int \frac{3}{x-1} dx = 3 \ln |x-1| + k$. Escrivint $\frac{2}{(x+1)^2} = 2(x+1)^{-2}$,

$$\int \frac{2}{(x+1)^2} dx = 2 \frac{(x+1)^{-1}}{-1} + k = -\frac{2}{x+1} + k.$$

b) (1,25 punts) Calcula $\int_2^4 \frac{3}{x-1} dx$ i $\int_0^1 \frac{2}{(x+1)^2} dx$.

Solució. $\int_2^4 \frac{3}{x-1} dx = [3 \ln |x-1|]_2^4 = 3 \ln 3 - 3 \ln 1 = 3 \ln 3$.

$$\int_0^1 \frac{2}{(x+1)^2} dx = \left[-\frac{2}{x+1} \right]_0^1 = -1 - (-2) = 1.$$